

Question ID: b07a7634

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Expression of Ideas	Rhetorical Synthesis	Hard

Question

While researching a topic, a student has taken the following notes:

- Digital Light Synthesis (DLS) is a form of additive manufacturing that utilizes light to rapidly cure liquid resin into high-quality, 3D objects.
- Step 1: Ultraviolet (UV) light images are projected up into a pool of liquid resin, where the object’s first layer takes shape.
- Step 2: The partially cured resin object is raised, leaving a thin space (a “dead zone”) beneath it for oxygen and liquid resin to flow through.
- Step 3: The UV light passes through the dead zone—maintaining the flow of resin—and partially cures additional layers of the object.
- Step 4: When the resin object is complete, it is baked in an oven to complete the curing.

The student wants to describe how DLS cures 3D objects. Which choice most effectively uses relevant information from the notes to accomplish this goal?

Answer

- A. DLS is a form of additive manufacturing that creates a “dead zone” in which UV light solidifies layer by layer before being baked in an oven, creating a high-quality, 3D object.
- B. DLS cures 3D objects by passing through a “dead zone,” adding layers to the object, then curing the object in an oven.
- C. In DLS, UV light images are projected into a liquid resin pool to cure a 3D object layer by layer; once solidified, the object is baked in an oven.
- D. In DLS, UV light is projected into layers of liquid resin until the resin solidifies and passes through a “dead zone,” wherein the curing is completed.

Correct Answer: C

Rationale

Choice C is the best answer. The sentence describes how DLS cures 3D objects, noting that UV light is projected into resin to cure the object in layers, after which the object is baked to complete the curing process.

Choice A is incorrect. While the sentence does describe some elements of the curing process, the description misrepresents information in the notes; this sentence suggests that the UV light, rather than the 3D object, is baked in an oven. Choice B is incorrect. While the sentence does describe some elements of the curing process, the description misrepresents information in the notes: the UV light, not DLS, passes through the dead zone. Choice D is incorrect. While the sentence does describe some elements of the curing process, the description misrepresents information in the notes. This sentence suggests that curing is complete when the already-solidified resin passes through the dead zone; rather, liquid resin passes through the dead zone and is solidified in layers, after which curing is completed in the oven.